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(G · ‡Ubkb 269)

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Ref. no: CRTC/CEE/ 20221002/S-163

SUB-SOIL INVESTIGATION REPORT

PROPOSED FOR 05 STORIED RESIDENTIAL BUILDING
AT PLOT # 39, BLOCK-B, CITY CORNER R/A, TUKERBAZAR, SYLHET
DAG # 2344, KHA # 3315, J.L- 83, MOUJA- GUPAL UTTAR

CLIENT

MD. SIRAJUL ISLAM

Verification Code: 457ZxySQ
Verify report at: crtc-cee-sust.com

CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET.

SAMPLE COLLECTED BY

THE CONFIDENT SOIL ENGINEERS
Mohine-52/1, Lamapara, Shibgonj, Sylhet.
Mob: 01718284057, 01979284057

Introduction

A reasonably accurate conception of the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study of the sub-surface elements in this regard would be necessary to select the proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing, and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Borehole no	Depth	Quantities
01	Disturbed sample	01	0-50 ft	10
02	Disturbed sample	02	0-40 ft	08
03	Disturbed sample	03	0-40 ft	08
04	Disturbed sample	04	0-40 ft	08
05	Disturbed sample	05	0-40 ft	08
06	-	-	-	-
07	-	-	-	-
08	-	-	-	-
09	-	-	-	-
10	-	-	-	-

Laboratory Classifications

The disturbed soil samples supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with borehole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, the Bearing Capacity of soil samples, including skin friction & end bearing, was calculated.

Bearing capacity for shallow foundation

The N value gives valuable information regarding the consistency of cohesive soil and relative density of cohesion less soil. These relationships are shown in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

<u>Bearing capacity for deep foundation</u>

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less soil	
$q_{ult} = 30N;$ $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square(Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	5	0.557	0.586	0.638	0.658	0.710	0.773	0.600
	5.0-10	10	1.176	1.237	1.348	1.389	1.500	1.632	1.267
BH02	2.5-5.0	5	0.557	0.586	0.638	0.658	0.710	0.773	0.600
	5.0-10	9	1.020	1.073	1.169	1.205	1.301	1.416	1.099
BH03	2.5-5.0	4	0.464	0.488	0.532	0.548	0.592	0.644	0.500
	5.0-10	8	0.944	0.993	1.082	1.115	1.204	1.310	1.017
BH04	2.5-5.0	5	0.534	0.561	0.612	0.630	0.681	0.741	0.575
	5.0-10	9	1.060	1.115	1.215	1.252	1.352	1.471	1.142
BH05	2.5-5.0	5	0.557	0.586	0.638	0.658	0.710	0.773	0.600
	5.0-10	10	1.176	1.237	1.348	1.389	1.500	1.632	1.267
BH06	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH07	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH08	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH09	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH10	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-

Table 8: The allowable skin friction for Precast piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.049	0.049	0.049	0.024	0.049	-	-	-	-	-
10	0.138	0.118	0.096	0.040	0.118	-	-	-	-	-
15	0.118	0.138	0.118	0.040	0.138	-	-	-	-	-
20	0.254	0.246	0.246	0.136	0.136	-	-	-	-	-
25	0.248	0.248	0.160	0.160	0.160	-	-	-	-	-
30	0.232	0.232	0.232	0.232	0.232	-	-	-	-	-
35	0.224	0.224	0.224	0.669	0.224	-	-	-	-	-
40	0.216	0.216	0.216	0.669	0.216	-	-	-	-	-
45	0.208	-	-	-	-	-	-	-	-	-
50	0.200	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable end bearing for Precast piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.450	0.450	0.450	4.800	0.450	-	-	-	-	-
10	1.350	1.125	0.900	8.000	1.125	-	-	-	-	-
15	1.125	1.350	1.125	8.000	1.350	-	-	-	-	-
20	4.275	3.375	3.375	27.200	27.200	-	-	-	-	-
25	49.600	49.600	32.000	32.000	32.000	-	-	-	-	-
30	46.400	46.400	46.400	46.400	46.400	-	-	-	-	-
35	44.800	44.800	44.800	11.250	44.800	-	-	-	-	-
40	43.200	43.200	43.200	11.250	43.200	-	-	-	-	-
45	41.600	-	-	-	-	-	-	-	-	-
50	40.000	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.032	0.032	0.032	0.016	0.032	-	-	-	-	-
10	0.090	0.077	0.062	0.026	0.077	-	-	-	-	-
15	0.077	0.090	0.077	0.026	0.090	-	-	-	-	-
20	0.165	0.160	0.160	0.088	0.088	-	-	-	-	-
25	0.161	0.161	0.104	0.104	0.104	-	-	-	-	-
30	0.151	0.151	0.151	0.151	0.151	-	-	-	-	-
35	0.146	0.146	0.146	0.435	0.146	-	-	-	-	-
40	0.140	0.140	0.140	0.435	0.140	-	-	-	-	-
45	0.135	-	-	-	-	-	-	-	-	-
50	0.130	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.203	0.203	0.203	2.160	0.203	-	-	-	-	-
10	0.608	0.506	0.405	3.600	0.506	-	-	-	-	-
15	0.506	0.608	0.506	3.600	0.608	-	-	-	-	-
20	1.924	1.519	1.519	12.240	12.240	-	-	-	-	-
25	22.320	22.320	14.400	14.400	14.400	-	-	-	-	-
30	20.880	20.880	20.880	20.880	20.880	-	-	-	-	-
35	20.160	20.160	20.160	5.063	20.160	-	-	-	-	-
40	19.440	19.440	19.440	5.063	19.440	-	-	-	-	-
45	18.720	-	-	-	-	-	-	-	-	-
50	18.000	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

Based on analysis and information provided by the owner, the following recommendations are suggested for the proposed five (05) storied residential buildings.

Soil profile scenario

The overall soil information of the investigated site is quite regular among the boreholes, and the soil formation is vertically consistent. There have been nearly two distinct types of drilling scenarios observed.

For type A (borehole no.: 01, 02, 03 &0 4)- brown soft clay with low bearing capacity exists up to 10 ft from the EGL among the boreholes. Although a layer of brown stiff clay with good bearing capacity was found at 20 ft depth of EGL in all boreholes except borehole no.:05, the layer did not go all the way down. Finally, gray fine sand with silt having good bearing capacity begins at 25 ft of EGL in all boreholes and continues thereafter.

For type B (borehole no.: 04)- gray loose sand with low bearing capacity was found roughly up to the top 15 ft from EGL in this borehole. After that, gray fine sand with silt having good bearing capacity has been observed from 20-30 ft depth below EGL. This borehole shows that the gray hard clay layer with good bearing capacity begins at 30 feet of EGL and extends to a greater depth in the borehole.

From the above discussion, it is apparent that the whole scenario of the proposed site is quite different in nature.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Shallow foundations may not be suitable for a five-story residential building because most boreholes contain soft clay soil with low bearing capacity at the top. Thus, the soft clay soil layer is highly susceptible to settlement.#The bearing capacity for the shallow foundation given in **table 7** is based on existing soil conditions.

However, deep foundation may be preferentially recommended to ensure the stability of the structure. The size and depth of the pile should be selected in light of the bearing capacity given in the report.

Safety provisions

It is strongly recommended to check the **settlement** of the site considering the building load while designing either a shallow or deep foundation. Obviously, **pile load test** covering the entire project area must be carried out to confirm the service pile's design parameters (*if pile foundation is considered*). Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work, including workmanship, should be considered for the safe design of a shallow or deep foundation.

Attachments

The layout of the boreholes of the investigated site and the bore-log profiles with SPT values are shown in the Appendix. The client performed no particular laboratory test for the proposed site.

Special notes

These suggestions are absolutely tentative, and the selection of type, depth, and dimensions of the foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **50** ft as the client provided.

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